Anika Kanwal

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Professional Summary

A highly motivated and research-focused Computer Science academic with over a decade of teaching experience across Australia and Pakistan, including roles at Swinburne University of Technology and The University of Adelaide. Holding a PhD in Computer Science from Swinburne University of Technology, my expertise spans deep learning, machine learning, predictive modelling, and intelligent systems, supported by publications in international journals and conferences. I have taught a broad range of undergraduate and postgraduate computing courses, including programming, artificial intelligence, data structures, software development, and business analytics. Recognised for inclusive and student-centred teaching practices, I was awarded the Vice-Chancellor's Reconciliation Award for empowering Indigenous perspectives in STEM. Technically proficient in Python, TensorFlow, PyTorch, and modern development tools, I bring a strong combination of academic expertise, research capability, and commitment to excellence in higher education.

Education

Doctor of Philosophy in Computer Science (Supervisor: Dr. Man Lau) — *March 2023*

Swinburne University of Technology — *Melbourne, Australia*

Thesis: Stock Price Prediction Models using Neural Networks

Masters of Computer Sciences — *March 2016*

Capital University of Science and Technology — *Islamabad, Pakistan*

Thesis: HMM based feature extraction from API calls for malware classification

Bachelors of Computer Sciences — *March 2013*

University of Poonch — *AJK, Pakistan*

Project: Expert Health Assistant

Experience

Swinburne University of Technology — *Aug. 2019 - PRESENT*

Tutoring — *Melbourne, Australia*

- Delivering face-to-face and online tutorials/lab demonstrations, problem-solving, grading for undergraduate and masters students in:
  (i) Introduction to programming (2019 - 2025)
  (ii) Technology Inquiry Project (2025)
  (iii) Data Structures (2022 - 2026)
  (iv) Technology in an Indigenous Context Project (2023 - 2026)
  (v) Object Oriented Programming (2024 - 2025)
  (vi) Web Development (2024)
  (vii) Introduction to Artificial Intelligence (2023)
  (vii) Business Analytics and Visualisation (2023, 2024)
  (ix) Technical Software Development (2019 - 2023)
  (x) Software Development for Mobile Devices (2022)
  (xi) Internet of things (2022)
  (xii) Computer Systems (2022)

Charles Strut University Melbourne Campus — *Feb. 2026 - PRESENT*

Sessional Lecturer - Unit Coordinator — *Melbourne, Australia*

- Delivering face-to-face Lectures for undergraduate and master's students in Programming Principles

Swinburne University of Technology — *Aug. 2024 - Sep. 2024*

Seminars — *Melbourne, Australia*

- Delivering online Seminars for undergraduate students in:
  (i) Technology Innovation Project

University of Adelaide Melbourne Campus
Jul. 2023 - Jul. 2024

Lecturer — *Melbourne, Australia*

- Delivering face-to-face Lectures for undergraduate and master's students in:
  (i) Introduction to Statistical Machine Learning
  (ii) Operating Systems
  (iii) Introduction to Computer Systems, Networks and Security
  (iv) Computer Systems
  (v) Artificial Intelligence
  (vi) Networking Fundamentals

Swinburne University of Technology
Aug. 2022 - Jun. 2023

Lecturing — *Melbourne, Australia*

- Delivering online Lectures for undergraduate students in:
  (i) Technical Software Development (Sem 1 and Sem 2 (2022, 2023))

Swinburne University of Technology
Jan. 2019 - Mar. 2023

PhD Candidate — *Melbourne, Australia*

- Independently conducting literature review on stock price prediction.
- Strong background in machine learning and deep learning, with hands-on experience in building and training Predictive and classification Models
- Proficiency in programming languages such as Python and relevant libraries (e.g., TensorFlow, PyTorch, scikit-learn)
- Developing CuDNNLSTM, CuDNNGRU and CNN based Hybrid Model
- Extensive comparisons among 15+ Predictive Model
- Writing academic reports and publications in LaTEX and MS Word.

Islamabad Model College for Girls, G9/2
Jan. 2018 - Dec. 2018

Senior Computer Teacher — *Islamabad, Pakistan*

- Delivering face-to-face lecture, lab demonstrations, problem-solving and grading of students of C++ Programming Language.

Rawalpindi Women University
Nov. 2016 - Jan. 2018

Lecturer — *Rawalpindi, Pakistan*

- Delivering face-to-face lecture, lab demonstrations, problem-solving grading for undergraduate in:
  (i) Programming Fundamentals
  (ii) Theory of Automata
  (iii)Information Systems
  (iv) Web Programming
  (v) Technical Software Development
  (vi) Object Oriented programming
- BSCS Co-ordinaator

Govt. Degree College for Women
Sept. 2014 - Sep. 2015

Lecturer — *Rawalpindi, Pakistan*

- Delivering face-to-face lectures, conducting lab demonstrations, solving problems, and grading for undergraduate students in Computer Science.

Ministry of Commerce, Government of Pakistan
Sep. 2013 - Sep. 2014

Internship — *Islamabad, Pakistan*

- PHP Developer
- Worked on Complain management system

EVS Professional Training Institute
Jan. 2013 - Jan. 2016

Administrator (Part-Time) — *Rawalpindi, Pakistan*

- Manage data in spreadsheets and reports
- Keep records and reports up to date
- Help maintain the budget plan
- Organize and schedule meetings and events
- Supervise other staff and delegate responsibilities
- Handle technical issues in their area of expertise

Project

Classification of COVID-19 X-ray Images

May 2022

Research Project

Melbourne, Australia

- This research received Australian Academy of Science Regional Collaborations Programme grant funding (Grant No. 11285136981).
- 2dCNN-BiCuDNNLSTM: Hybrid Deep-Learning-Based Approach for Classification of COVID-19 X-ray Images

Publications

- Anika Kanwal, Man Lau, et al. (2022). "BiCuDNNLSTM-1dCNN—A hybrid deep learning-based predictive model for stock price prediction". In: Expert Systems with Applications https://doi.org/10.1016/j.eswa.2022.117123
- Anika Kanwal, Siva Chandrasekaran, et al. (2022). "2dCNN-BiCuDNNLSTM: Hybrid Deep-Learning-Based Approach for Classification of COVID-19 X-ray Images". In: Sustainability https://doi.org/10.3390/su14116785
- Anika Kanwal, Siva Chandrasekaran, et al. (2023). "MGRU-M1dCNN: A Hybrid Model for stock price Prediction". In: 2023 15th International Conference on Computer and Automation Engineering (ICCAE 2023) held during March 3-5, 2023. https://doi.org/10.1109/ICCAE56788.2023.10111233
- Anika Kanwal, Siva Chandrasekaran and Adnan Akhunzada (2023). "Stock Price Prediction using CuDNNLSTM and multiple CNN layers". In: 2023 IEEE International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings) https://doi.org/10.1109/AIBThings58340.2023.10291016.
- Anika Kanwal, Man Lau, et al. (2024). "Navigating the Stock Market: The Role of State-of-the-Art Deep Learning Techniques". In: Al-Balqa Journal for Research and Studies. https://doi.org/10.35875/vvf5k957
- Anika Kanwal, Siva Chandrasekaran and Adnan Akhunzada (2025). "A Hybrid Attention-BiGRU Model for Accurate Stock Market Prediction". In: 2025 IEEE 4th International Conference on Computing and Machine Intelligence (ICMI) https://doi.org/10.1109/ICMI65310.2025.11141054.
- Anika Kanwal, Man Lau, et al. (2026). "BiCuDNN(SLSTM-GRU)-1dCNN Predictive Model for Stock Price Prediction". UNDER-REVIEW

Awards

Vice-Chancellor's Reconciliation Award, 2023

Swinburne University of Technology, Australia

Vice-Chancellor's Reconciliation Award was granted in:

Dec, 2023

- Empowering Indigenous Australians knowledge for first-year STEM students to connect People with Technology

Silver Medal, 2016

Islamabad, Pakistan

Silver Medal was granted in:

Dec, 2016

- Master of Computer Science

Dean's Honor Roll, 2015

Islamabad, Pakistan

Dean's Honor Roll was granted in:

Spring, 2015

- Master of Computer Science

Excellence Award, 2014

AJK, Pakistan

Excellence Award was granted in:

June, 2014

- Bachelors of Computer Science

References

Dr Afzaal Azeem Chowdhary (Convenor)

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Lecturer

X

- School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, Australia
- achowdhary@swin.edu.au

Dr Man Lau (PhD supervisor)

X

Senior Lecturer

X

- School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, Australia
- elau@swin.edu.au

Dr Deepa Prabhu

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CERC Postdoctoral Research Fellow

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- Commonwealth Scientific and Industrial Research Organization (CSIRO) Sydney, Australia
- deepa.prabhu@csiro.au